Fiscal Year:	FY 2012	**Task Last Updated:**	FY 03/15/2012
PI Name:	Garofalo, Roberto M.D.		
Project Title:	Effect of Space Flight on Innate Immunity to Respiratory Viral Infections		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	77555-0366	**Congressional District:**	14
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	05/01/2010	**End Date:**	04/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-133 ; ISS NOTE: Extended to 4/30/2014 per NSSC information (Ed., 1/31/13) NOTE: Extended to 4/30/2013 per A. Chu/ARC (Ed., 3/6/12)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Casola, Antonella (University of Texas Medical Branch)		
Grant/Contract No.:	NNX10AI50G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our overarching hypothesis is that a combination of several factors from microgravity to stress, affect the human immune system during space flight, resulting in an impaired antiviral response in the respiratory tract. Three specific hypotheses that will be tested in this proposal are: 1. Space flight or ground conditions such as antiorthostatic suspension leads to an altered host response against two of the most common human respiratory pathogens, Respiratory Syncytial Virus (RSV) and Human metapneumovirus (hMPV); 2. This “deficient” antiviral response is characterized by a defect in both innate immunity, including IFN type I production in the respiratory tract, and T cell-mediated immune response; 3. The central events that cause this defect in antiviral immunity are the impaired trafficking and function of pDC in the respiratory mucosa.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space Applications: Space shuttle Discovery will bring NASA scientists one step closer to helping astronauts and the public discover ways to battle and prevent serious illness and infection. These investigations are expected to generate new scientific evidence of the immune pathways that are affected in antiviral host response during space flight and possible preventive or therapeutic approaches applicable to future space missions. Earth Applications: Understanding the function of the immune system during space flight may have great relevance to our understanding of the process of aging and/or stress-related immune-modulation (adjustments in the level of an immune response) on Earth. In particular, studying the innate host response against pathogens during/after space flight will provide novel data on the function of the respiratory mucosal response to viral pathogens. The goal of the Mouse Immunology-2 experiment is to discover what triggers and leads to an increased susceptibility to an infection. These findings can be used to help treat and prevent future astronauts from getting sick, as well as protect people with more vulnerable immune systems here on Earth, such as the elderly or young children.
	Space flight is a unique stress model impacted consistently or intermittently by physical stress, high G forces at the time of launch and landing, increased radiation, sleep deprivation, microgravity, and nutritional factors. This multitude of factors alters the immune system and could lead to compromised defenses against infections and tumors. Over the past two to three decades, there have been a large number of investigations of space effects on the immune system, and have been reported that space flight has profound immunosuppressive effects on humans and animals. These results raise the possibility that space flight could result in decreased resistance to infection in animals. Respiratory Syncytial Virus (RSV): Human respiratory syncytial virus (RSV) is the leading cause of severe lower respiratory tract infection in infants and a major cause of morbidity and mortality in elderly and immunocompromised adults. Approximately two-thirds of infants are infected with RSV during the first year of life, and 90% have been infected one or more times by 2 years of age. However, both infants and adults are susceptible to recurrent infections. In the United States, RSV infection is the most common cause of respiratory disease leading to hospitalization in children, costing an estimated $300–$400 million annually. In the USA alone, RSV is responsible for 70 000 to 126 000 infant hospitalization for pneumonia or bronchiolitis every year (http://www.who.int). The elderly also are at risk for severe RSV disease and 14 000 to 62 000 RSV-associated hospitalizations of the elderly occur annually in the USA. The fight against RSV is considered by the WHO as a priority. In 2005, it was estimated that 66 000–199 000 children younger than 5 years died from RSV associated acute lower respiratory infection, with 99% of these deaths occurring in developing countries. Worldwide, RSV is associated with an annual mortality rate of 160,000-600,000 deaths. RSV has been the focus of numerous efforts to develop a vaccine. To date, there is no safe and effective vaccine available; treatment options against RSV infection is limited. The main focus of this study has been to establish and characterize the effect of space flight on innate immune response to RSV infection in BALB/c mouse model. The pathogenesis of RSV lower respiratory tract infection is not completely understood, but it is likely to be mediated both by the direct effects of the virus and by the host’s inflammatory response. In the present study female BALB/c mice were flown on the final mission of Space Shuttle Discovery (STS-133) for 12.8 days and infected immediately after landing with respiratory syncytial virus (RSV) or saline. Since space flight influences immunological parameters the effect of space flight on the clinical disease parameters, viral replication, lung inflammation, BAL cellular infiltrate, production of cytokines, chemokines, type I IFN, and whole blood characteristics were explored herein. Our study could yield extensive information on the mechanism of the effect of space flight followed by RSV infection on immune response. This type of study could give us information that would be useful for developing countermeasures to help treat and prevent future astronauts from getting sick, as well as protect people with more vulnerable immune systems here on Earth, such as the elderly or young children. 1. Our results show that FLT mice after return to Earth weighed 8 and 12% less compared with AEM and VIV controls, respectively. This is in agreement with several other studies reporting a loss in body mass of rats and mice after space flight. The low body mass in our FLT mice may well be related to dehydration, less food consumption due to the main effect of the flight condition. 2. In our study, infection of BALB/c mice by intranasal inoculation with RSV produced observable symptoms of disease such as weight loss, and ruffled fur for all groups VIV, AEM, and FLT. RSV infection was associated with similar degree of body weight loss in all groups, but disease severity score was higher in the FLT group. The results from body mass for mock infected groups (VIV, AEM, and FLT) shows no significant differences during the experimental period. The body mass increased for mice from FLT mock infected ones after flight. 3. Consistent with increased RSV infection, the cellular population in BAL fluid was altered. RSV infection induces lung recruitment of neutrophils, which are significantly higher for the FLT group at day 1 post-infection compared with VIV and AEM ones, and decreased over the next time points. In addition, FLT mock infected animals revealed that neutrophils represent 45% from the main population at day 1 post-infection compared with VIV and AEM mock infected ones where macrophages were the main population. 4. In our study, cultures of homogenized lungs, demonstrated the presence of RSV at day 5 post-infection for all groups VIV, AEM, and FLT. In addition, there was a trend suggesting an increase in RSV loads, which was more prominent in mice from the FLT group compared with AEM and VIV ones. 5. In this study, mouse lung tissue was investigated to evaluate a potentially health hazard effect of space flight, since

Task Progress:

the lungs are very vulnerable to increased levels of radiation, inhalation of bacteria, and low ambient oxygen. We found that examination of the lungs of RSV infected mice at day 7 post-infection revealed similar pathological changes for all groups VIV, AEM, and FLT manifested by multilayered perivascular and peribronchial inflammatory infiltrates, with monocytes and occasional neutrophils invading the vascular bed, and associated mild focal necrosis of vascular endothelial cells. The pathological changes in the lungs appeared slightly milder in AEM group compared with VIV and FLT ones. In contrast, no histopathological changes were observed in mock inoculated mice from all groups.

6. Several experiments in space flight during the last decade showed significant changes in blood parameters. Neutrophils are important elements for astronaut health, and increases in the number of neutrophils of more than 100% for crewmembers have been reported after space flight. The number of neutrophils is highest immediately after landing. Studies conducted on the first 41 U.S. space shuttle astronauts demonstrated that the number of lymphocytes and eosinophils in the peripheral blood decreased after flight.

The results of the current study demonstrate that space flight followed by RSV infection can affect the blood parameters. An increase in neutrophils response in blood at post flight, day 1 post-infection was observed in our study for RSV infected FLT animals compared with ground controls (VIV, and AEM). The observed increased in neutrophils response for RSV infected animals was transitory and the values returned to the control levels for all groups (VIV, AEM, and FLT) for the next points of the experiment (day 5, and 7). We observed a decreased in lymphocytes response for RSV infected FLT animals compared with ground controls at day 1 post-infection, which started to increased at day 5 and 7 post-infection.

The data also showed that RBCs number was elevated at day 1 post-infection in the FLT RSV infected group compared with the VIV ones. The pattern of changes in RBCs at day 1 post-infection was similar to those of both the HGB and HCT for the RSV infected groups (FLT vs. VIV). The post flight values showed no significant differences in peripheral blood parameters for mock infected animals from all groups (VIV, AEM, and FLT), however our samples were collected at 18 hrs post-infection and not immediately after landing.

In contrast to rapid changes in blood neutrophils, the increase in bronchoalveolar lavage (BAL) neutrophils in RSV infected animals is detectable about 5 days after inoculation. In addition, differential counts of bronchoalveolar cells revealed that neutrophils represented up to 45% of cells in the FLT mock-infected mice compared with AEM and VIV mock groups in which macrophages were the main population (95%) at day 1 post-infection, whether no significance changes in neutrophils response in peripheral blood in mock infected animals was observed in all groups (VIV, AEM, and FLT) during the experiment period.

7. In recent years it has become clear that cytokines and chemokines involved in host response to RSV infection are thought to play a central role in the pathogenesis. The profile, relative abundance and kinetics of cytokines that are produced in the airways in the course of viral respiratory infections are all factors that contribute to shaping host immunity, inflammatory response, and ultimately viral shedding. RSV stimulates airway epithelial cells and residents leukocytes to release cytokines. Epithelial cells are key regulators of the innate immune response against viral mediators in response to RSV infection. These include cytokines (interleukin-1,-6), tumor necrosis factor (TNF-a), several chemokines (KC, macrophage inflammatory protein (MIP)-1a, monocyte chemotactic protein (MCP-1), RANTES), type I IFN (IFN-a/ß), and growth factors (GM-CSF, G-CSF). Macrophages play a crucial role in respiratory viral infections. In our model, BAL samples from RSV-infected mice from FLT group had significantly greater concentrations of the CC chemokines CCL2 (MCP-1) at 18h post-infection compared to those from RSV-infected ground control groups (VIV, and AEM), but returned to control levels on day 5 and 7 post-infection. While monocyte chemotactic protein (MCP) production occurs in the context of virus infection, the mechanisms underlying this event are not well understood. MCP-1 (CC-chemokine ligand 2, CCL-2) belongs to the CC-chemokine family, binds specifically to CCR2, and is important for monocyte, macrophage, and T-cell recruitment in several inflammatory models.

In our study, immunomodulatory cytokine IL-17 remained elevated in RSV-infected FLT group on days 5 and 7 of infection compared with ground control groups. IL-17 is a CD4+ T cell-derived cytokine that has been previously shown to play a role in the immune response to several different lung infections. Additionally, it has been shown that IL-17 is upregulated in respiratory syncytial virus (RSV) infection. The proinflammatory cytokines IL-1a, IL-1ß, TNF-a, and CC chemokines CCL3 (MIP-1a) were differentially induced by RSV infection in VIV, AEM, and FLT groups at day 1 post-infection. Overall, BAL concentrations of IL-1a, IL-1ß, TNF-a, and MIP-1a were significantly greater in RSV infected mice from VIV group compared with FLT infected ones. Production for MIP-1a has been previously reported in RSV-infected mice, and plays a critical role in the pathogenesis of RSV disease both in the mouse model and in naturally acquired infections. At early stage of the infection immunomodulatory cytokines, including IL-12 (p70), IL-12 (p40), were also differentially induced by RSV VIV ground control group compared with FLT ones. On the other hand, at later time points IL-12 (p40) was induced in significantly higher amounts by RSV, with a peak of production around day 7 of infection. IL-12 (p40) is produced primarily by cells of macrophage and dendritic cell lineage. The role of IL-12 (p40) in the context of RSV infection, particularly in view of the relatively high concentrations of the cytokine that we recovered in BAL, is currently unclear. IL-12 (p40) is known to antagonize IL-12 (p70) activity and to trigger IFN-gamma production by NK and T cells.

In addition, our results show that production of GM-CSF and G-CSF in BAL samples peaked at 18 h after RSV infection and by day 5 had reached near-constitutive levels, with no significant differences between groups (VIV, AEM, and FLT).

Overall, BAL samples from RSV infected mice had high levels of interleukin IL-6, IL-9, IL-10, IL-13, and chemokines, MIP-1ß, RANTES at 18 h post-infection in all groups, and there is a rapid decline for the next time points. The CXC chemokine KC was induced in higher amounts in RSV infected mice from all groups (VIV, AEM, FLT) at 18 h post-infection. KC, a mouse IL-8 homologue, is a potent chemoattractant for neutrophils, the predominant lung inflammatory cell during acute RSV infections.

Our study demonstrated that RSV induces significantly increased serum concentrations of chemokines MCP-1, MIP-1ß, RANTES, tumor necrosis factor TNF-a, and granulocyte colony-stimulating factor G-CSF in FLT group compared with ground controls, VIV, and AEM at day 1 post-infection. Our results show that granulocyte-macrophage colony-stimulating factor (GM-CSF) is elevated in FLT mock infected group at day 1 post-infection. RSV infection induced high levels of GM-CSF at day 1 post-infection for all groups (VIV, AEM, and FLT), however peaked at day 7 only for FLT RSV infected group. We understand that one of the problems in examining inflammatory cytokines in serum is that these measurements may not reflect accurately the response in airways. However, the immunological

responses have previously been studied in serum and plasma in children with respiratory infection with RSV.

As previously reported by others and us, space flight has been shown to alter many immunological parameters, among these affected are the production of cytokines. Studies in both human and animals models of space flight alter the production of cytokines (interferon production, responsiveness of bone marrow cell granulocytes/monocyte-colony stimulating factor). This suggests that there are selective effects of space flight on immune responses, and not all cytokines and chemokines are affected in the same way by space flight.

8. Type I IFNs, among the first anti-viral agents, were discovered because of their anti-viral activity against influenza virus and seen to be central in the early anti-viral response. The type I IFNs synthesis during RSV infection may be associated closely with severity of the disease. Many different factors seem to determine the production of type I IFN by RSV, including cell type and viral strain. In addition, the kinetics of type I IFNs produce in RSV-infected mice or patients is different from other respiratory virus. In humans infected with RSV, these studies are difficult to interpret considering the nature of the samples analyzed (nasal washes, nasal aspirates and sera), variations in subject age, timing of sample isolation and IFN measurement techniques. The fact that all of these studies demonstrated the presence of type I IFN in the nasopharyngeal samples suggests that type I IFN plays an important but transient role during RSV infection. As previously reported by others and us, RSV can induce high levels of type I IFNs immediately upon infection, but this burst was short-lived with a sharp decline in type I IFNs production as the infection progressed.

Our study provides evidence that in RSV-infected animals from all groups (VIV, AEM, and FLT), IFN-a/ß production peaked at 18h post-infection, but type I IFN activity decreased sharply bellow limit of detection over the next two time points (day 5, and 7). The concentrations of IFN-a/ß in BAL were significantly higher at 18h post-infection in RSV-infected mice from the FLT group compared with the ground controls (VIV, and AEM), after short-term space flight exposure of the animals. Furthermore, we observed that concentration of IFN-a in serum is significantly higher for the FLT RSV infected group at day 1 post-infection compared with VIV and AEM infected ones. Altogether, these data demonstrate that space flight can induce high levels of type I IFNs immediately upon RSV infection, but there is a rapid decline in IFN-a/ß production for the next time point of experiment.

RSV pathogenesis is a complex process dependent on the interaction of viral and host determinants. These studies are critical for developing new therapeutic strategies against RSV infections. The data reported in this study are the results of a rare opportunity, in which we have been able to show in the BALB/c model that short-term space flight followed by RSV infection induced significant changes in parameters that can influence immune defense mechanisms, changes that may become significant during long-term space missions. Most importantly, animals models have proved invaluable for caring out infectious disease experiments and have increased our understanding of the biological significance of the effect of space flight conditions on the immune system. Understanding the effect of space flight on the immune response is essential to ensure the health, safety, and performance of space station crew members.

ACCOMPLISHMENTS POST-FLIGHT EXPERIMENTS

In the last year, 2011, of this project we made considerable progress towards fulfilling our primary aims, as previously described above.

The following posters were presented in 2011, and 2 manuscripts will be submitted for publication in 2012:

1. “EFFECT OF SPACE FLIGHT ON INNATE IMMUNITY TO RESPIRATORY VIRAL INFECTIONS” - Post-doctoral week apreciation meeting, UTMB Galveston, TX, October 2011 – Teodora Ivanciuc was awarded 1st place. Teodora Ivanciuc,1 Deepthi Kolli,1 Paula Dumars,2 Elena Sbrana,3 Yashoda Hosakote-Madaiah,1 Zinzi Blell,4 Lan Pang,1 Antonella Casola,1,4 Nicole Rayl,5 , and Roberto P. Garofalo,1,4*. 1 Division of Clinical and Experimental Immunology & Infectious Diseases (CEIID), Dept. of Pediatrics, UTMB Galveston, 2 Lockheed Martin Engineering & Science Solutions, NASA Ames Research Center , 3 Dept. of Microbiology, Dept. of Pathology - Autopsy Service, UTMB Galveston, 4 Dept. of Microbiology and Immunology, UTMB Galveston, 5 NASA Ames Research Center

2. “EFFECT OF SPACE FLIGHT ON INNATE IMMUNITY TO RESPIRATORY VIRAL INFECTIONS I: CLINICAL DISEASE PARAMETERS” - ASGSB meeting, San Jose, CA, November 2-6, 2011. Teodora Ivanciuc,1 Deepthi Kolli,1 Paula Dumars,2 Elena Sbrana,3 Yashoda Hosakote-Madaiah,1 Zinzi Blell,4 Lan Pang,1 Antonella Casola,1,4 Nicole Rayl,5 , and Roberto P. Garofalo,1,4*. 1 Division of Clinical and Experimental Immunology & Infectious Diseases (CEIID), Dept. of Pediatrics, UTMB Galveston, 2 Lockheed Martin Engineering & Science Solutions, NASA Ames Research Center , 3 Dept. of Microbiology, Dept. of Pathology - Autopsy Service, UTMB Galveston, 4 Dept. of Microbiology and Immunology, UTMB Galveston, 5 NASA Ames Research Center

3. “EFFECT OF SPACE FLIGHT ON INNATE IMMUNITY TO RESPIRATORY VIRAL INFECTIONS II: CYTOKINE and BLOOD PARAMETERS ”- ASGSB meeting, San Jose CA, November 2-6, 2011. Teodora Ivanciuc,1 Deepthi Kolli,1 Paula Dumars,2 Yashoda Hosakote-Madaiah,1 Zinzi Blell,4 Lan Pang,1 Antonella Casola,1,3 Nicole Rayl,4 , and Roberto P. Garofalo,1,3*. 1 Division of Clinical and Experimental Immunology & Infectious Diseases (CEIID), Dept. of Pediatrics, UTMB Galveston, 2 Lockheed Martin Engineering & Science Solutions, NASA Ames Research Center , 3 Dept. of Microbiology and Immunology, UTMB Galveston, 4 NASA Ames Research Center

4. “EFFECT OF SPACE FLIGHT ON INNATE IMMUNITY TO RESPIRATORY VIRAL INFECTIONS I: CLINICAL DISEASE PARAMETERS” - manuscript to be submitted to American Journal of Physiology - Lung Cellular and Molecular Physiology. Teodora Ivanciuc,1 Deepthi Kolli,1 Paula Dumars,2 Elena Sbrana,3 Yashoda Hosakote-Madaiah,1 Zinzi Blell,4 Lan Pang,1 Antonella Casola,1,4 Nicole Rayl,5 , and Roberto P. Garofalo,1,4*. 1 Division of Clinical and Experimental Immunology & Infectious Diseases (CEIID), Dept. of Pediatrics, UTMB Galveston, 2 Lockheed Martin Engineering & Science Solutions, NASA Ames Research Center , 3 Dept. of Microbiology, Dept. of Pathology - Autopsy Service, UTMB Galveston, 4 Dept. of Microbiology and Immunology, UTMB Galveston, 5 NASA Ames Research Center

5. “EFFECT OF SPACE FLIGHT ON INNATE IMMUNITY TO RESPIRATORY VIRAL INFECTIONS II: CYTOKINE and BLOOD PARAMETERS ” - manuscript to be submitted to Journal of Applied Physiology Teodora Ivanciuc,1 Deepthi Kolli,1 Paula Dumars,2 Yashoda Hosakote-Madaiah,1 Zinzi Blell,4 Lan Pang,1 Antonella Casola,1,3 Nicole Rayl,4 , and Roberto P. Garofalo,1,3*. 1 Division of Clinical and Experimental Immunology & Infectious Diseases (CEIID), Dept. of Pediatrics, UTMB Galveston, 2 Lockheed Martin Engineering & Science Solutions, NASA Ames Research Center , 3 Dept. of Microbiology and Immunology, UTMB Galveston, 4 NASA Ames Research Center

Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Ivanciuc T, Kolli D, Dumars P, Sbrana E, Hosakote-Madaiah Y, Blell Z, Pang L, Casola A, Rayl N, Garofalo RP. "Effect of space flight on innate immunity to respiratory viral infections." Post-doctoral week appreciation meeting, UTMB Galveston, TX, September 19-23, 2011. Post doctoral week appreciation meeting, UTMB Galveston, TX, September 19-23, 2011. , Sep-2011
Abstracts for Journals and Proceedings	Ivanciuc T, Kolli D, Dumars P, Sbrana E, Hosakote-Madaiah Y, Blell Z, Pang L, Casola A, Rayl N, Garofalo RP. "Effect of space flight on innate immunity to respiratory viral infections I: clinical disease parameters." 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 99. , Nov-2011
Abstracts for Journals and Proceedings	Ivanciuc T, Kolli D, Dumars P, Sbrana E, Hosakote-Madaiah Y, Blell Z, Pang L, Casola A, Rayl N, Garofalo RP. "Effect of space flight on innate immunity to respiratory viral infections II: cytokine and blood parameters." 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 99. , Nov-2011
Awards	Ivanciuc T. "Teodora Ivanciuc was awarded 1st place Overall Grand Prize for 'Effect of space flight on innate immunity to respiratory viral infections' at the Post-doctoral week appreciation meeting, UTMB Galveston, TX, September 19-23, 2011." Sep-2011